Department of Computer Science & Engineering
The University of Texas at Arlington

Architectural Design Specification
CSE 4316: Senior Design I
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EGG UTA
(Encapsulated Gardening Grounds)

David Denny
Samuel Horn
Kaitlin Martin
Chanuthi Subasignhage
Khoi Tran

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Contents

1 Introduction **5**

2 System Overview **6**
2.1 Sensor Layer Description 6
2.2 Desktop Application and Software Layer Description 6
2.3 Physical Materials and Build Layer Description 7
2.4 Hardware and Automation Layer Description 7

3 Subsystem Definitions & Data Flow **9**

4 Sensor Layer Subsystems **10**
4.1 Moisture Sensor 10
4.2 Nutrition Sensor 11
4.3 Humidity and Temperature Sensors 12

5 Desktop Application and Software Layer Subsystems **13**
5.1 Data Storage 13
5.2 Auto/Manual Modal Control and User Interface 14
5.3 Data Processing and Analysis 15

6 Physical Materials and Build Layer Subsystems **17**
6.1 Water Reservoir and Drain Pan 17
6.2 Base (Materials and Assembly) 18
6.3 Cover (Materials and Assembly 19
6.4 Plumbing 20

7 Hardware and Automation Layer Subsystems **22**
7.1 Microcontroller 22
7.2 Watering (Automation) 23
7.3 Fans (Automation) 24

List of Figures

1 Architectural Layers for the EGG and its associated Desktop Application 6
2 Data Flow for the EGG and its associated Desktop Application 9
3 Moisture sensor subsystem diagram 10
4 Nutrition sensor subsystem diagram 11
5 Humidity/temperature subsystem diagram 12
6 Data Storage subsystem diagram 13
7 Auto/Manual Modal Control and User Interface subsystem diagram 14
8 Data Processing and Analysis subsystem diagram 15
9 Water Reservoir and Drain Pan subsystem diagram 17
10 Base Materials and Assembly subsystem diagram 18
11 Cover Materials and Assembly subsystem diagram 19
12 Plumbing subsystem diagram 20
13 Microcontroller subsystem diagram 22
14 Watering Automation subsystem diagram 23
15 Fans Automation subsystem diagram 24

List of Tables

1 Moisture sensor subsystem interfaces 11
2 Nutrition sensor subsystem interfaces 11
3 Humidity/temperature sensor subsystem interfaces 12
4 Data Storage subsystem interfaces 14
5 Auto/Manual Mode Control and User Interface subsystem interfaces 15
6 Data Processing and Analysis subsystem interfaces 16
7 Water Reservoir and Drain Pan subsystem interfaces 18
8 Plumbing subsystem interfaces 21
9 Microcontroller subsystem interfaces 23
10 Watering Automation subsystem interfaces 24
11 Fan Automation subsystem interfaces 25

1 Introduction

The Encapsulated Gardening Grounds, otherwise known as the EGG, is a project and product designed to assist individuals and small-scale growers maintaining crops more efficiently through a combination of an embedded system. This system is predicated on requisite hardware and software to assist these individuals, including various sensors and an associated desktop application to offer maintenance advice and manual control over various hardware automation.

More specifically, the EGG system comprises a controlled, enclosed environment optimized for plant growth, supported by an array of sensors and an automated irrigation and humidity control system. These features are tied to a companion desktop application that provides users with real-time environmental data about the EGG's current characteristics, maintenance advice, and direct control over automated systems.

At its core, the EGG aims to enable users to become more self-sufficient by giving them the tools needed to monitor and maintain their crops effectively in varied environments, including urban and indoor areas. Users will benefit from data driven insights and advice tailored to these specific environments, allowing for an easier experience taking care of crops.

This introduction captures the broad, general requirements and goals of this project and product. Contained in this document are the various Architectural elements of the project needed to implement it during its development. This includes the broad, higher-level layers of each main system in the EGG, and their requisite subsystems. Additionally, the interfaces between each system is listed in each system's description, outlining the various ways in which the embedded system will interact with each other to accomplish the goals and requirements of the product.

2 System Overview

The overall structure of the EGG system includes four distinct layers: Sensors, Desktop Application and Software, Physical Materials and Build, and Hardware and Automations. The purpose of these layers is to address both the software and hardware features of the EGG whilst also accounting for data gathering and physical make-up of the EGG. The EGG needs to be able to read and store data, process user input, and physically house and attend to the plants using various embedded tools.

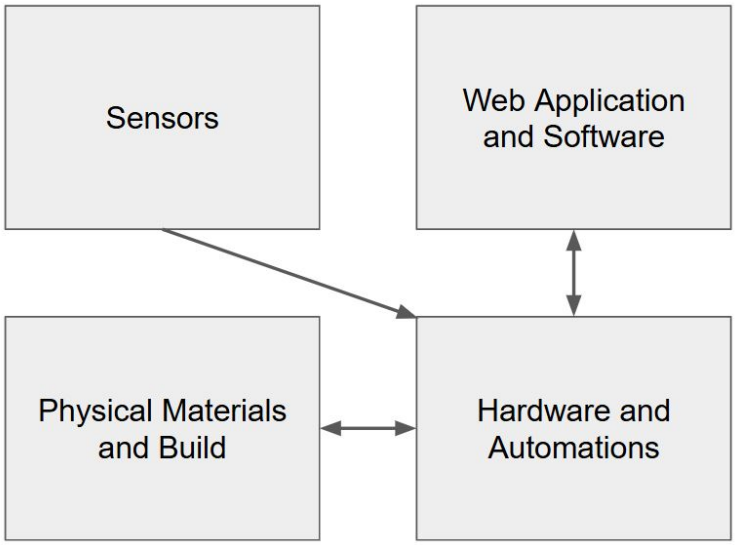

Figure 1: Architectural Layers for the EGG and its associated Desktop Application

2.1 Sensor Layer Description

The purpose of the Sensor Layer is to collect data from the EGG environment in real time. Since sensors cannot process or act upon the data, this layer's data flow goes directly to the Microcontroller specified in the Hardware and Automations layer (this is further explained in section 3: Subsystem Definitions and Data Flow). The various sensors and their functions are as follows:

- Moisture Sensor: detects soil moisture by sending an electric current through the prongs, which reads resistance with a voltage comparator (LM393 chip). If the value is above or below the expected set value, the output will indicate "too high" or "too low".
- Nutrition Sensor: detects nitrogen, phosphorus, and potassium levels in the soil to indicate the strength and growth of the plant. This is done in a similar fashion: a probe is used to read the conductivity of the soil after an electric pulse is sent (JXCT Soil NPK).
- Humidity / Temperature Sensor: detects the overall humidity and temp of the EGG environment. This sensor must be carefully placed above the soil to read the moisture in the air. This will likely be accomplished using a DHT22 sensor. Depending on the plant, the output of this sensor may cause the Microcontroller to activate the sprinkler system to raise the humidity if necessary.

2.2 Desktop Application and Software Layer Description

The purpose of the Desktop Application and Software Layer is to provide a Graphic User Interface to the user of the EGG in which they can monitor the plant's overall health and take command of the EGG's manual features. This layer is a vital bridge of communication between the EGG's data and the user,

where they can be alerted to changes in the environment's temperature, humidity, soil nutrition, and moisture. The functions of this layer are as follows:

- Read from a Database (likely MongoDB) that holds the sensor data, and process the data. To accomplish this, there needs to be a separate Database used as the plant library that contains the specifications of several plant species that the user will be able to specify. The desktop app will then be able to compare the read values with the expected values and give a proper analysis on the differences between them.
- Recommend steps to take for the user. After data processing, the desktop app needs to display its findings to the user and give suggestions on what could be improved for the plant to be healthy, such as more water, less sunlight, etc.
- Provide controls over manual functions like the fan, the sprinklers, or the water drip. These features should also have an "automatic" mode, where the user allows the EGG environment to independently activate these tools when needed based on data analysis.
- Display overall status of the plant. The desktop app should always display basic information as well as a log of what decisions have been made, like when the fan was last activated, how long it ran, etc.

2.3 Physical Materials and Build Layer Description

The purpose of the Physical Materials and Build Layer is to provide a shelter that will house the plants as they grow. This layer was made distinct from the Hardware and Automations Layer due to the fact that the features of this layer are the basic qualities required for any garden, not just a "smart" garden. The functions of this layer are as follows:

- A source of water: the EGG will utilize a multipurpose drain pain as both a water reservoir and a collection point for excess water that filters through the soil. This water must be replaced by the user periodically to ensure the water is free of contaminants, which will build as it is being reused.
- A bed of soil: the base of the EGG will provide the space needed to fill with soil in a grid system. This base will sit directly above the drain pan, which will have the ability to slide out from underneath the EGG like a drawer.
- An enclosed covering: the plants will be safely enclosed by a sliding lid, which can optionally be left open or closed. This is to ensure the portability of the EGG system as well as a controlled environment safe from outside threats such as pests, harsh weather, or thievery.
- Waste control and maintenance: while the EGG can handle excess water, it is important to have the EGG's features easily accessible for other forms of maintenance. For example, the user will be required to harvest the plants, refresh the soil, and sanitize the plumbing after long periods of time to prevent contaminant build up. Most parts should be completely visible and/or detachable, like the tubes used for the water pump.

2.4 Hardware and Automation Layer Description

The purpose of the Hardware and Automation Layer is to access the various devices used in the EGG's controlled environment and to process commands made by the User Interface. The Microcontroller is like the EGG's CPU: it sends and receives information from the user, and translates user requests into commands for the various tools like the fan, the sprinkler system, or the water drip. The functions of this layer are as follows:

- Populate the EGG's Database with sensor information: all of the readings need to be collected and sent to the Database so that it may be accessed and analyzed by the desktop application.
- Receive requests from the User Interface: these requests range from manual control of the EGG's devices, to commands to resume automated control. In the latter case, the Microcontroller must be able to maintain communication with the desktops app for instructions.
- Detect the various devices: the Microcontroller should alert the user if communication has been lost between any of the sensors or the other tools.

3 Subsystem Definitions & Data Flow

The following diagram displays the various subsystems that make up each layer of the Encapsulated Gardening Grounds' Architecture. Additionally, the numbered connections show how each subsystem interacts with each other, which mirrors the above layer diagram's connections on a more specific level. Most importantly, many of the subsystems from the sensors and physical layers directly interact with the Microcontroller subsystem within the hardware layer. Additionally, the Software and Desktop Application interacts with the Microcontroller to gather data and transmit control signals to the Hardware's automated subsystems.

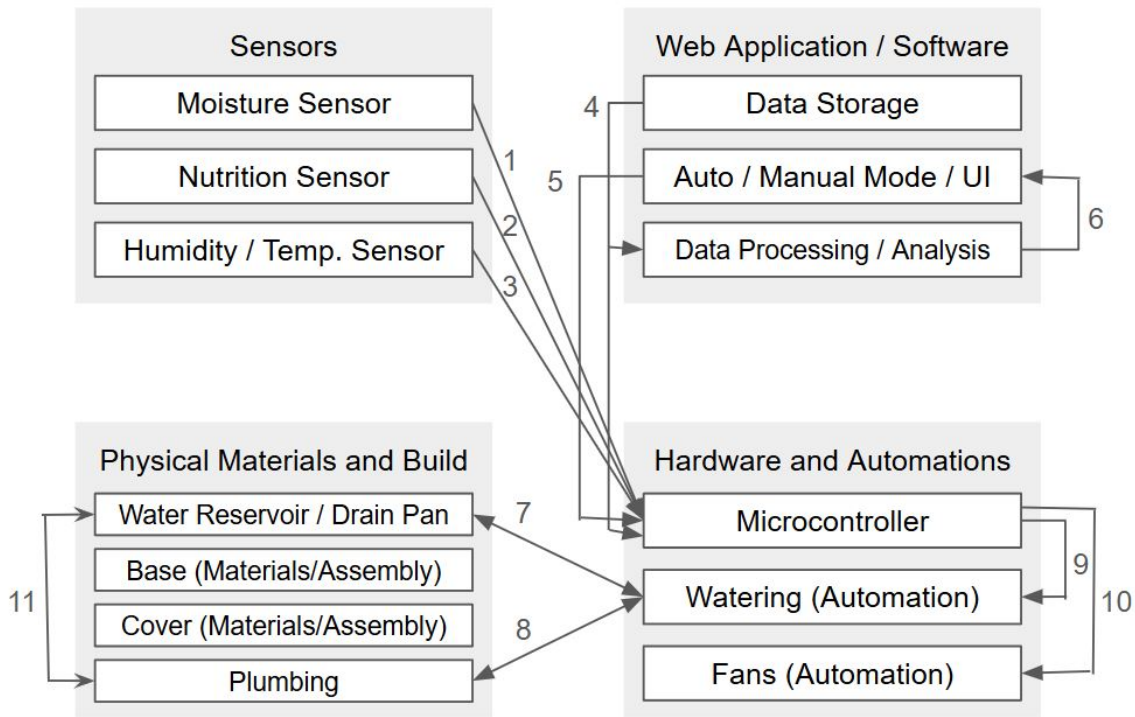

Figure 2: Data Flow for the EGG and its associated Desktop Application

4 Sensor Layer Subsystems

The sensor layer consists of three types of sensors: moisture, nutrition, and humidity/temperature. These sensors collect data about the EGG environment and send it to the microcontroller.

4.1 Moisture Sensor

The EGG will use the LM393 hygrometer to measure the moisture of the enclosure's soil and relay it to the microcontroller.

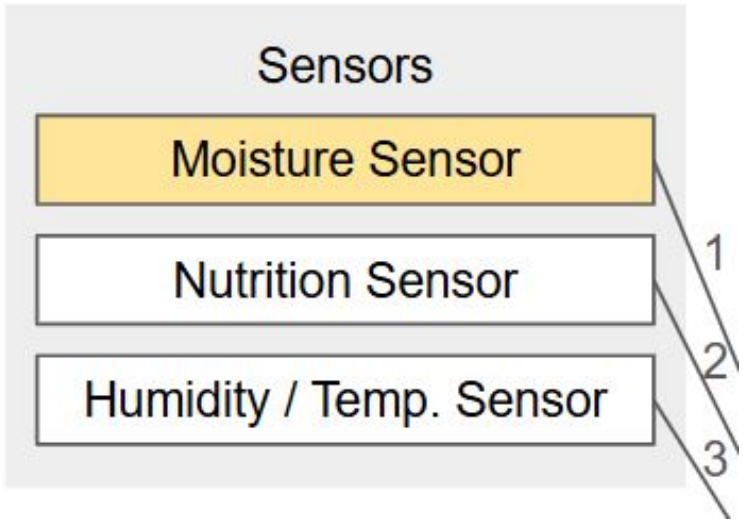

Figure 3: Moisture sensor subsystem diagram

4.1.1 Assumptions

- The waterproof and non-waterproof components of the sensor should be distinctly separated.
- The sensor should be located such that it receives data from the relevant parts of the soil.
- The integrity of the sent data is maintained from collection to retrieval by the microcontroller.

4.1.2 Responsibilities

- The sensor must accurately measure the moisture of the soil to allow automated watering. The LM393 uses analog output, where the measured voltage reflects the level of moisture in the soil, as well as digital output, where the sensor sends binary information based on whether the soil moisture is higher or lower than a predetermined value. Either or both will be used to accomplish this goal.
- The sensor should be robust enough to work as intended for a long period of time before eventual wear and replacement.

4.1.3 Subsystem Interfaces

Table 1: Moisture sensor subsystem interfaces

ID	Description	Inputs	Outputs
#1	Interface with EGG microcontroller	Real soil moisture	Analog/digital soil moisture data

4.2 Nutrition Sensor

A nutrition sensor will collect data concerning the health of the soil. It will likely be an NPK sensor (measuring nitrogen, phosphorus, and potassium), such as the JXCT Soil NPK sensor.

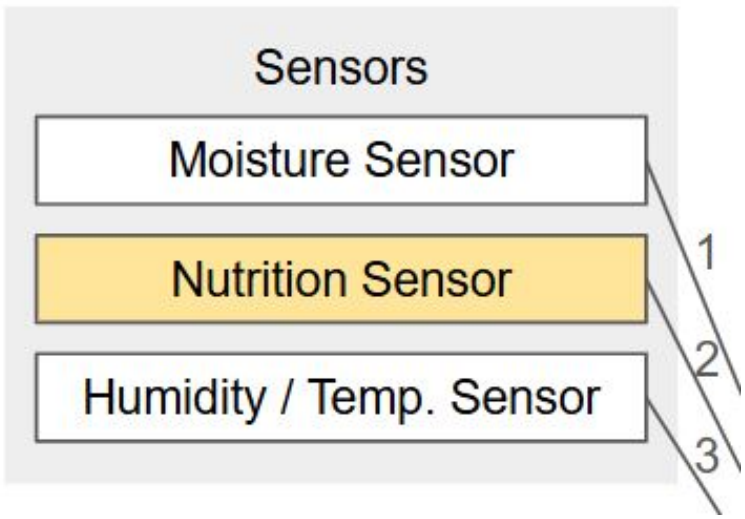

Figure 4: Nutrition sensor subsystem diagram

4.2.1 Assumptions

- The integrity of the sent data is maintained from collection to retrieval by the microcontroller.

4.2.2 Responsibilities

- The sensor should relay accurate soil nutrition information to the microcontroller.
- The sensor should be robust enough to work as intended for a long period of time before eventual wear and replacement.

4.2.3 Subsystem Interfaces

Table 2: Nutrition sensor subsystem interfaces

ID	Description	Inputs	Outputs
#2	Interface with microcontroller	Real soil nutrtion	Analog soil nutrtition data

4.3 Humidity and Temperature Sensors

In order to measure both the temperature and humidity of the EGG environment, we will be using the DHT22 sensor.

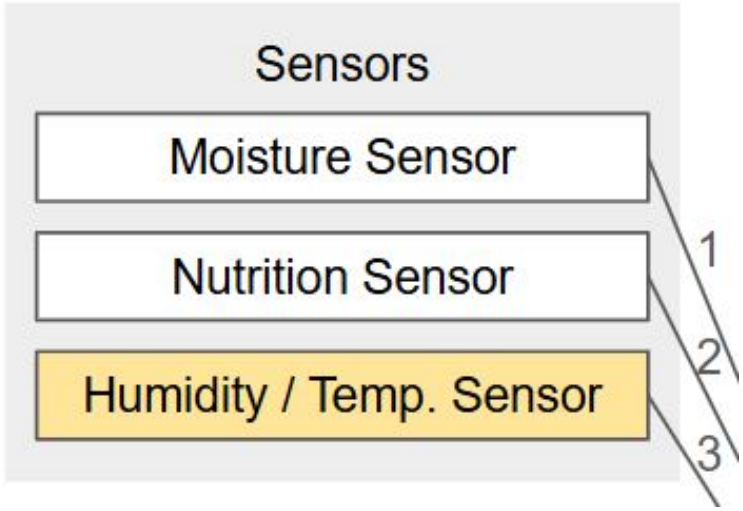

Figure 5: Humidity/temperature subsystem diagram

4.3.1 Assumptions

- The sensor must be located such that other components do not interfere with its operation, such as ones that give off heat, or spread water.
- The integrity of the sent data is maintained from collection to retrieval by the microcontroller.

4.3.2 Responsibilities

- The sensor must collect accurate temperature and humidity data about the EGG environment.
- The sensor should be robust enough to work as intended for a long period of time before eventual wear and replacement.

4.3.3 Subsystem Interfaces

Table 3: Humidity/temperature sensor subsystem interfaces

ID	Description	Inputs	Outputs
#3	Interface with microcontroller	Real humidity Real temperature	Analog humidity and temperature data

5 Desktop Application and Software Layer Subsystems

The Web Application and Software Layer serves as the interface between the user and the underlying physical hardware of the automated plant grower system. This layer is responsible for managing user interactions, storing, and processing sensor data, as well as providing feedback through a user-friendly interface.

5.1 Data Storage

This subsystem is responsible for storing data, user preferences, and system state information. It is a shared resource for other subsystems within the application and software layer, particularly for the Auto/Manual Mode UI and the Data Processing/Analysis module.

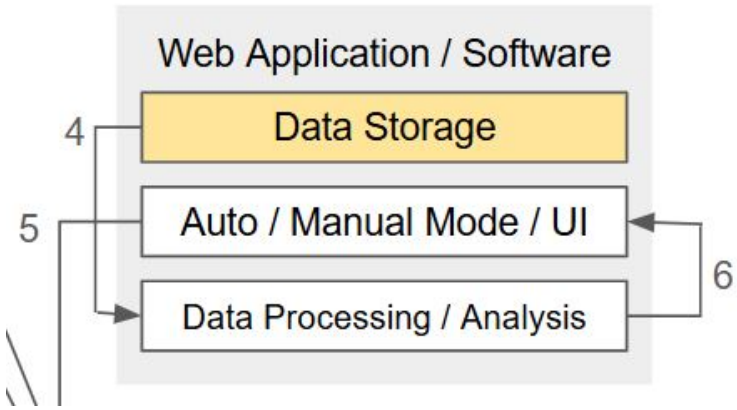

Figure 6: Data Storage subsystem diagram

5.1.1 Assumptions

The following assumptions are to be considered for the Data storage subsystem:

- Assumes consistent format for incoming sensor data (e.g., JSON with fields for temperature, humidity, soil moisture, etc)
- The desktop application has network access to cloud-based or local storage solutions.
- code will be stored and run on SD card on the microcontroller.

5.1.2 Responsibilities

The following responsibilities are satisfied by the Data Storage subsystem:

- Store raw and processed sensor data (e.g., temperature, humidity, moisture levels).
- Save user-configured settings.
- Log events such as water cycles and alerts.
- Provide query functionality for data visualization and trend analysis.
- Allow other subsystems to access or modify stored data as needed.

5.1.3 Subsystem Interfaces

The following table shows the various ways in which the EGG Hardware interacts with other subsystems in the project:

Table 4: Data Storage subsystem interfaces

ID	Description	Inputs	Outputs
#01	Interface with Auto/Manual Mode UI	User settings Mode toggles	Current settings last saved preferences
#02	Interface with Data Processing/Analysis	Raw sensor data	Stored processed data
#03	External data input	Sensor readings	Confirmation of storage

5.2 Auto/Manual Modal Control and User Interface

The Auto/Manual Modal Control and UI subsystem is the primary means by which users interact with the automated plant grower system. It allows users to switch between automatic and manual control modes, configure system preferences, and view real-time system status and environmental data. This subsystem acts as a bridge between the user and the rest of the desktop application and software layer.

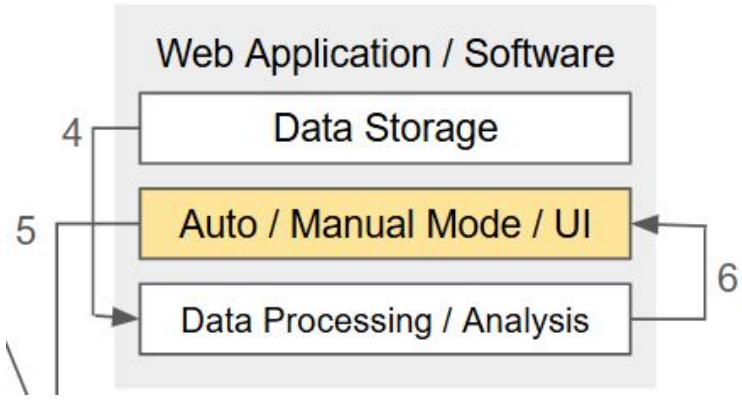

Figure 7: Auto/Manual Modal Control and User Interface subsystem diagram

5.2.1 Assumptions

The following assumptions are to be considered for the The Auto/Manual Modal Control and UI subsystem:

- The user interacts with the system via mobile device.
- The UI is built with a modern web framework (e.g., React Native, React.js , Tailwind CSS) and exchanges data with backend services through API requests, which connect to a MongoDB database.
- Real-time system updates (sensor reading, alerts) are made available to the UI through Data Processing/Analysis subsystem.

5.2.2 Responsibilities

The following responsibilities are satisfied by the Auto/Manual Modal Control and UI subsystem:

- Provide user interface for selecting control mode (Auto/Manual).
- Display real-time sensor data and system status updates.
- Allow users to configure system parameters (e.g., watering thresholds).
- Reflect alerts or automated decisions made by the Data Processing subsystem.
- Synchronize current state and preferences with Data Storage.

5.2.3 Subsystem Interfaces

The following table shows the various ways in which the EGG Hardware interacts with other subsystems in the project:

Table 5: Auto/Manual Mode Control and User Interface subsystem interfaces

ID	Description	Inputs	Outputs
#01	Interface with Auto/Manual Mode UI	User settings Mode toggles	Current settings last saved preferences
#02	Interface with Data Processing/Analysis	Raw sensor data	Stored processed data
#03	External data input	Sensor readings	Confirmation of storage

5.3 Data Processing and Analysis

The Data Processing and Analysis subsystem interprets raw sensor data from the automated plant grower's hardware. It will determine environmental trends, assess thresholds, and generate meaningful insights that are then used by both the Data Storage and Auto/Manual Mode UI subsystems. This component supports automating decisions and alerting users to system anomalies and required maintenance.

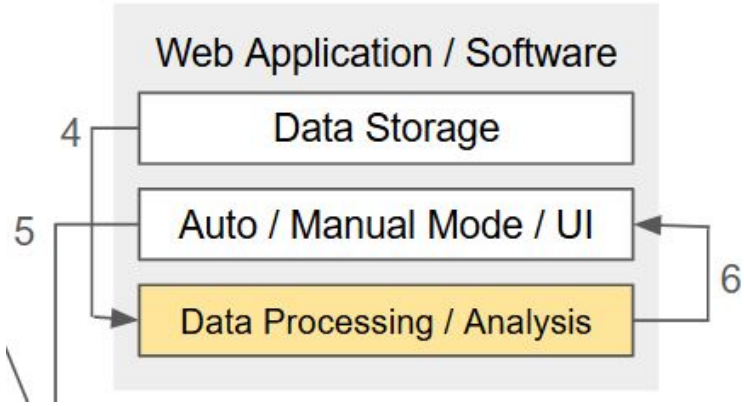

Figure 8: Data Processing and Analysis subsystem diagram

5.3.1 Assumptions

The following assumptions are to be considered for the Data Processing/Analysis subsystem:

- Incoming data from sensors is already timestamped and in a standard format
- Logic for threshold checks is configured by our team and is fetched from the Data Storage
- Processed data is used to trigger UI updates or automated action.

5.3.2 Responsibilities

The following responsibilities are satisfied by the Data Processing/Analusis subsystem:

- Process incoming raw sensor data.
- Detect threshold violations or abnormalities.
- Trigger actions or flags for auto-watering.
- Communicate processed results to the storage subsystem for logging and to the UI subsystem for visualization.

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5.3.3 Subsystem Interfaces

The following table shows the various ways in which the EGG Hardware interacts with other subsystems in the project:

Table 6: Data Processing and Analysis subsystem interfaces

ID	Description	Inputs	Outputs
#01	Interface with Data Processing and Analysis subsystem	Processed sensor data	Visualization updates, status indicators
#02	Interface with Data Storage subsystem	Saved user preferences, current system mode	Mode change requests, updated user settings

6 Physical Materials and Build Layer Subsystems

This section of the ADS details the subsystems that support the structural construction or mechanical operation of the device. Structural elements support the device's longevity and contain the encapsulated environment. Mechanical elements support plant maintenance systems, such as plumbing and water supply systems.

6.1 Water Reservoir and Drain Pan

This subsystem details the EGG's water supply system, which consists of the reservoir and drain pan. The water reservoir provides water to the EGG's irrigation system, where a pump draws water into the plumbing network, which will be detailed later. After irrigation, any unused water is captured as runoff by the drain pan and returned to the reservoir. Ideally, the drain pan will employ some sort of filtering process that will stop soil or other debris from returning to the reservoir. This will allow the device to conserve water efficiently while protecting the plumbing system and keeping maintenance simple. The user will also be able to refill the reservoir manually as needed. The materials used for this subsystem should be nonreactive and durable for water use and the external environment.

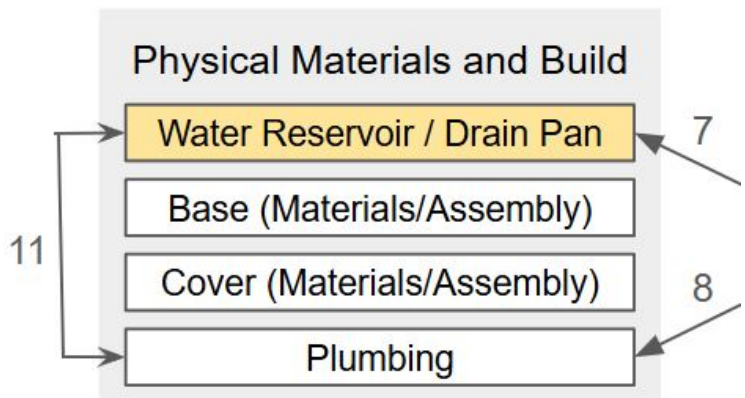

Figure 9: Water Reservoir and Drain Pan subsystem diagram

6.1.1 Assumptions

The following assumptions are to be considered for the Watering Automation:

- The user will refill the reservoir with water when the water supply gets low.
- The user will maintain and clean the device routinely to prevent clogging or molding.
- The water put into the device is clear of pollutants and debris.

6.1.2 Responsibilities

The following responsibilities are satisfied by the water reservoir and drain pan:

- The water reservoir is responsible for containing the water that the EGG's plumbing will direct to other irrigation and other processes.
- The drain pan is responsible for reducing the EGG's water consumption by recycling used water.
- The drain pan is responsible for preventing pollutants from entering the EGG's water systems.

6.1.3 Subsystem Interfaces

The following table shows the various ways in which the water reservoir and drain pan interact with the other subsystems in the project:

Table 7: Water Reservoir and Drain Pan subsystem interfaces

ID	Description	Inputs	Outputs
#7	Interface with Watering Systems	Excess water from plants	N/A
#11	Interface with plumbing	N/A	Water for irrigation systems

6.2 Base (Materials and Assembly)

This subsystem contains the device components and gardening elements. Together with the cover subsystem, it encapsulates the gardening environment, separating the internal climate from external conditions. The materials used for this subsystem should be nonreactive and durable, suitable for exposure to water and outdoor environments.

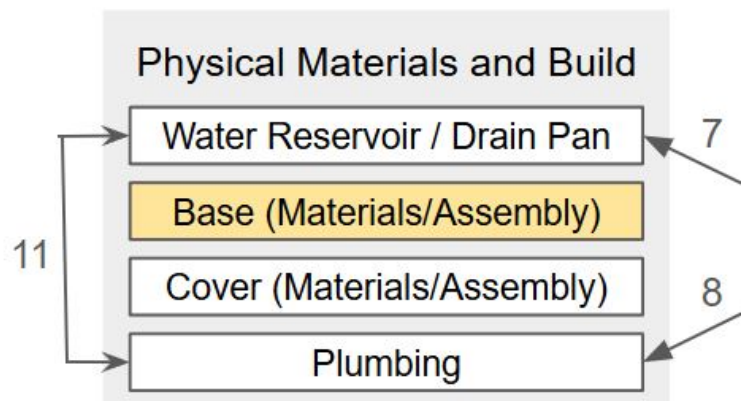

Figure 10: Base Materials and Assembly subsystem diagram

6.2.1 Assumptions

The following assumptions are to be considered for the base:

- The user will maintain and clean the device routinely to prevent clogging or molding.
- The structural materials used are corrosion-resistant and chemically inert relative to soil, water, and fertilizer.

6.2.2 Responsibilities

The following responsibilities are satisfied by the base:

- The base is responsible for separating and protecting the internal gardening environment from external weather conditions.
- The base is responsible for housing all of the internal components without wearing significantly over time.

6.3 Cover (Materials and Assembly

This subsystem contains the device components and gardening elements. Together with the cover subsystem, it encapsulates the gardening environment, separating the internal climate from the external conditions. The materials used for this subsystem should be nonreactive and durable, suitable for exposure to water and outdoor environments. The cover will allow the user to see into the EGG while closed and act as a lid, allowing the user to open and access the internal components, both electronic and organic.

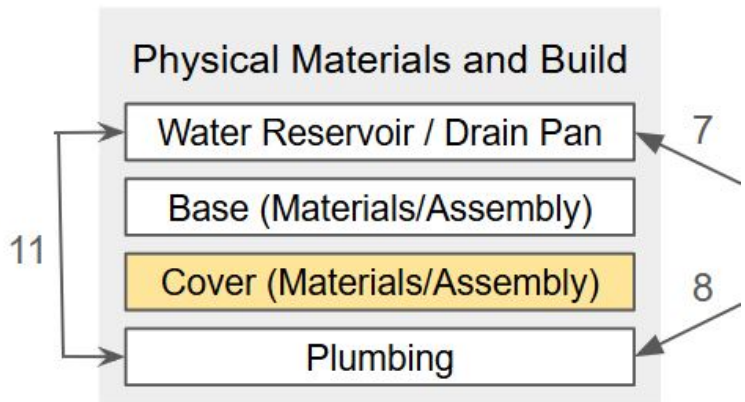

Figure 11: Cover Materials and Assembly subsystem diagram

6.3.1 Assumptions

The following assumptions are to be considered for the cover:

- The user will maintain and clean the device routinely to prevent clogging or molding.
- The structural materials used are corrosion-resistant and chemically inert relative to soil, water, and fertilizer.
- The materials used will not degrade due to UV, rain, and temperature.

6.3.2 Responsibilities

The following responsibilities are satisfied by the cover:

- The cover is responsible for separating and protecting the internal gardening environment from external weather conditions.
- The cover is responsible for housing all of the internal components without wearing significantly over time.
- The cover must allow the user to access the internal components, such as the electronics and plants.
- The cover must allow the user to see into the device while closed.

6.4 Plumbing

The Plumbing subsystem enables the delivery of water from the reservoir to the plants and ensures safe transport of excess water to the drain pan. Plumbing consists of tubing, pumps, and connectors designed to handle low-pressure water flow. The materials used for this subsystem should be nonreactive and durable for water use and the external environment.

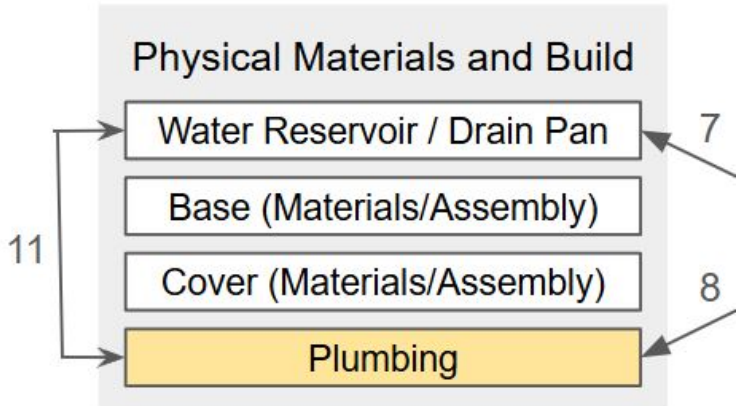

Figure 12: Plumbing subsystem diagram

6.4.1 Assumptions

- The water that flows through the plumbing is relatively clean and free of large particles.
- The user will routinely check and maintain the plumbing system to ensure it is working properly and is free of clogs, kinks, leaks, or other issues that may effect the device's performance.
- The plumbing layout will minimize leaks, kinks, and blockages.
- The pump and plumbing materials will not be effected by temperature and weather.
- The water pump capacity matches system size and watering needs.

6.4.2 Responsibilities

- The plumbing is responsible for transporting water from the reservoir to the watering systems based on sensor inputs or manual commands.
- The plumbing is responsible for ensuring proper drainage of excess water to the drain pan.
- The plumbing is responsible for protecting the electronic components from water damage.
- The plumbing is responsible for staying durable and watertight over a long period of time.

6.4.3 Subsystem Interfaces

Table 8: Plumbing subsystem interfaces

ID	Description	Inputs	Outputs
#8	Interface with watering systems	N/A	Water from the reservoir
#11	Interface with the water reservoir and drain pan	Water for irrigation systems	N/A

7 Hardware and Automation Layer Subsystems

This portion of the ADS describes the various hardware feature that intend to be used in the Encapsulated Gardening Ground's design. Each of these peripherals implement a function or feature towards one of the EGG's overall goals. The hardware includes the primary microcontroller, watering automation items, and fan automation items. It should be noted that the group of sensors used in the EGG's design is excluded from this Layer to distinguish between separate types of hardware utilized.

7.1 Microcontroller

The Microcontroller serves as the primary data collector and sender for the EGG. The controller is directly connected to the various sensors in order to drive and sample them as needed. Additionally, the controller is to drive the EGG's automated systems, including the pump and watering system, and the fans for dehumidifying. The controller will directly interact with the data storage system to keep data to be worked on and output by the associated web application.The selected controller for the EGG is Arduino UNO R4 WiFi.

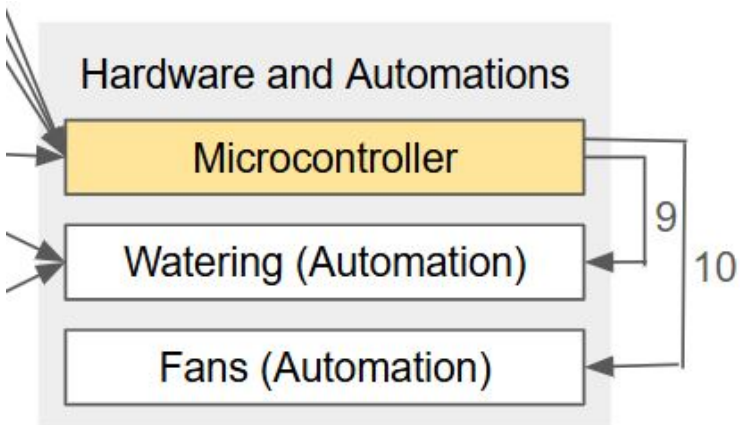

Figure 13: Microcontroller subsystem diagram

7.1.1 Assumptions

The following assumptions are to be considered for the Microcontroller:

- The controller can interact with the data storage at a requisite speed, determined by the web application. This means the controller will be able to speed fast enough to operate on. This data transfer may occur over WiFi, over local storage, or both.
- The controller will have enough ports to account for the three primary sensors used and the data interactions.

7.1.2 Responsibilities

The following responsibilities are satisfied by the Microcontroller:

- The controller is responsible for transmitting data sampled from the sensors to the data storage implementation.
- The controller is responsible for driving the watering and fan automated features.

7.1.3 Subsystem Interfaces

The following table shows the various ways in which the EGG Hardware interacts with other subsystems in the project:

Table 9: Microcontroller subsystem interfaces

ID	Description	Inputs	Outputs
#1	Moisture Sensor Sampling	Moisture Sensor	N/A
#2	Nutrition Sensor Sampling	Nutrition Sensor	N/A
#3	Humidity/Temp Sensor Sampling	Humidity/Temp Sensor	N/A
#5	Automation and Manual Input	Auto/Manual Mode/UI	N/A
#9	Watering Automation Interfacing	N/A	Watering Automation
#10	Fan Automation Interfacing	N/A	Fan Automation

7.2 Watering (Automation)

The Watering Automation encompasses the hardware needed to drive the watering of the crop held within the Encapsulated Gardening Grounds. Primarily, this includes the Microcontroller's ability to drive the pump, allowing for the pumping of water from the basin/drain pan to the EGG's watering outputs. This process can either be operated fully automatically, or manually operated.

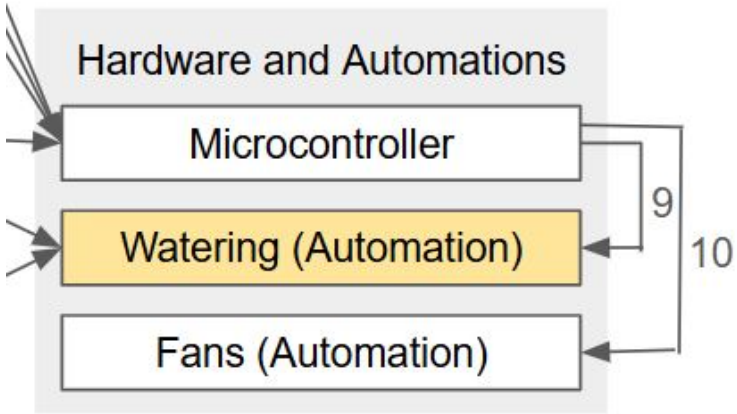

Figure 14: Watering Automation subsystem diagram

7.2.1 Assumptions

The following assumptions are to be considered for the Watering Automation:

- Data inputs to the Microcontroller, that in turn drive the watering system, will be successfully passed to the Microcontroller. This is to ensure that valid data is given to the automated watering system.
- All associated hardware (plumbing and the water basin) will be available for utilization when testing the watering automations.

7.2.2 Responsibilities

The following responsibilities are satisfied by the Automated Watering:

- The watering system is responsible for taking in driving outputs from the microcontroller to drive the pump and other needed hardware for watering automation.
- The pump and other peripherals in the watering automation must take these automatic and manual signals both wired and wirelessly.

7.2.3 Subsystem Interfaces

The following table shows the various ways in which the EGG Hardware interacts with other subsystems in the project:

Table 10: Watering Automation subsystem interfaces

ID	Description	Inputs	Outputs
#7	Water Tank/Drain Pan Interfacing	Water Reservoir / Drain Pan	Watering Automation
#8	Plumbing System Interfacing	Plumbing	Watering Automation
#9	Microcontroller Interfacing / Driving	Microcontroller	N/A

7.3 Fans (Automation)

The Fan Automation Systems describes the most simple hardware component in the Encapsulated Gardening Grounds. For the purposes of temperature and humidity control, fans are placed within the EGG, and run as needed to facilitate new oxygen, lower temperatures, and dehumidify the encapsulated environment. The fan(s) are simply driven by the Microcontroller when needed.

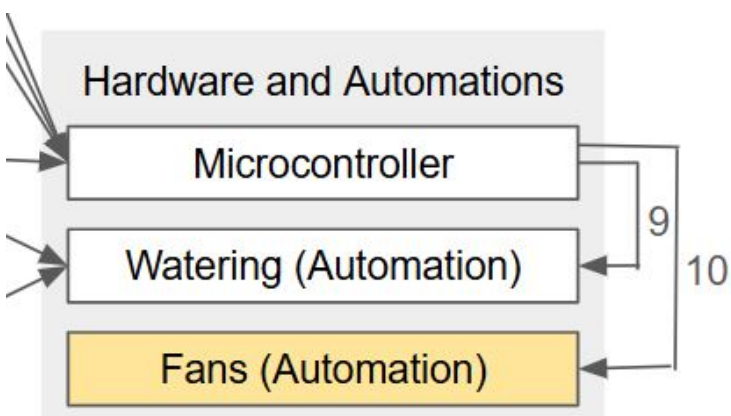

Figure 15: Fans Automation subsystem diagram

7.3.1 Assumptions

The following assumptions are to be considered for the Watering Automation:

- Data inputs to the Microcontroller, that in turn drive the fans, will be successfully passed to the Microcontroller. This is to ensure that valid data is given to the automated fan system.

7.3.2 Responsibilities

The following responsibilities are satisfied by the Automated Watering:

- The watering system is responsible for taking in driving outputs from the microcontroller to drive the fans for automations, controlling temperature and humidity.

7.3.3 Subsystem Interfaces

The following table shows the various ways in which the EGG Hardware interacts with other subsystems in the project:

Table 11: Fan Automation subsystem interfaces

ID	Description	Inputs	Outputs
#10	Microcontroller Interfacing / Driving	Microcontroller	N/A